

LATISH CHARPE

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OBJECTIVE

Data Science and AI undergraduate (dual-enrolled at MIT and IIT Madras) with hands-on experience in Python, SQL, and Power BI, developing real-time dashboards, AI applications, and data-driven solutions. Strong foundation in statistics, data visualization, and machine learning, with a passion for solving real-world problems through technology and continuous learning.

EDUCATION

B.Tech in Artificial Intelligence & Data Science

2023 – 2027

Maharashtra Institute of Technology, Chhatrapati Sambhajinagar

BS in Management & Data Science

2024 – Present

Indian Institute of Technology, Madras

TECHNICAL SKILLS

- **Programming & Analysis:** Python, SQL, Data Analysis, Pandas, NumPy, Fast API, React
- **Visualization & Reporting:** Power BI, Matplotlib, Seaborn, Microsoft Excel
- **Tools & Platforms:** GitHub, Google Cloud, Jupyter Notebook, Render, Vercel, Streamlit

PROJECTS

Car Price Prediction & Estimation System

- Developed an ML-based car price prediction system using a 15,241-record CarDekho dataset.
- Performed data cleaning, preprocessing, feature handling, and categorical encoding.
- Compared Linear Regression, Decision Tree, Random Forest, and Gradient Boosting, achieving $R^2 \approx 0.72$ with Random Forest.
- Built an interactive Streamlit dashboard for price estimation and PDF report generation.

Tech Stack: Python, Pandas, NumPy, Scikit-learn, Streamlit, Joblib.

AI-Based Industrial Plastic Defect Detection System

- Developed a computer vision system using YOLOv11m to automatically detect and localize 16 types of defects in injection-molded plastic components.
- Built an AI-powered inspection dashboard with image upload, bounding-box visualization, confidence scores, defect identification, and possible cause analysis.
- Implemented a rule-based process correction system that provides recommended injection-molding adjustments; achieved 63.50% mAP@0.50 ϕ on a fixed clean test set.

Tech Stack: Python, YOLOv11m, PyTorch, FastAPI, OpenCV, React.

IPL Match Analysis Dashboard

- Analyzed 500+ IPL matches across 10+ seasons to uncover team performance trends, win percentages, and player statistics.
- Built interactive visuals highlighting top run-scorers, strike rates, and head-to-head team comparisons for data-driven cricket insights.
- Designed a clean dashboard layout enabling stakeholders to filter by season, team, or player in real time.

Tech Stack: Power BI, Excel, SQL.

ACHIEVEMENTS & CERTIFICATIONS

- **Gen AI Exchange Hackathon(Hack2Skill)** — Shortlisted for Top 100 team in the a GenAI Exchange hackathon.
- **Smart India Hackathon (2025)** — Shortlisted in the Internal hackathon round.
- **Data Analysis with Python (IBM Developer)**
- **Data Analytics Using Power BI (Tech Tip 24)**
- **Artificial Intelligence in Industrial & Management Engineering (NPTEL IIT Kanpur)**
- **Introduction to Machine Learning(NPTEL IIT Kharagpur)**
- **Python for Data Science(NPTEL IIT Madras)**